ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CNTT & TT

BÁO CÁO BÀI TẬP
HOME WORK 1

Họ tên: Nguyễn Đức Toàn

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Mã lớp: 166155

Hà Nội, Ngày 4 tháng 3 năm 2026

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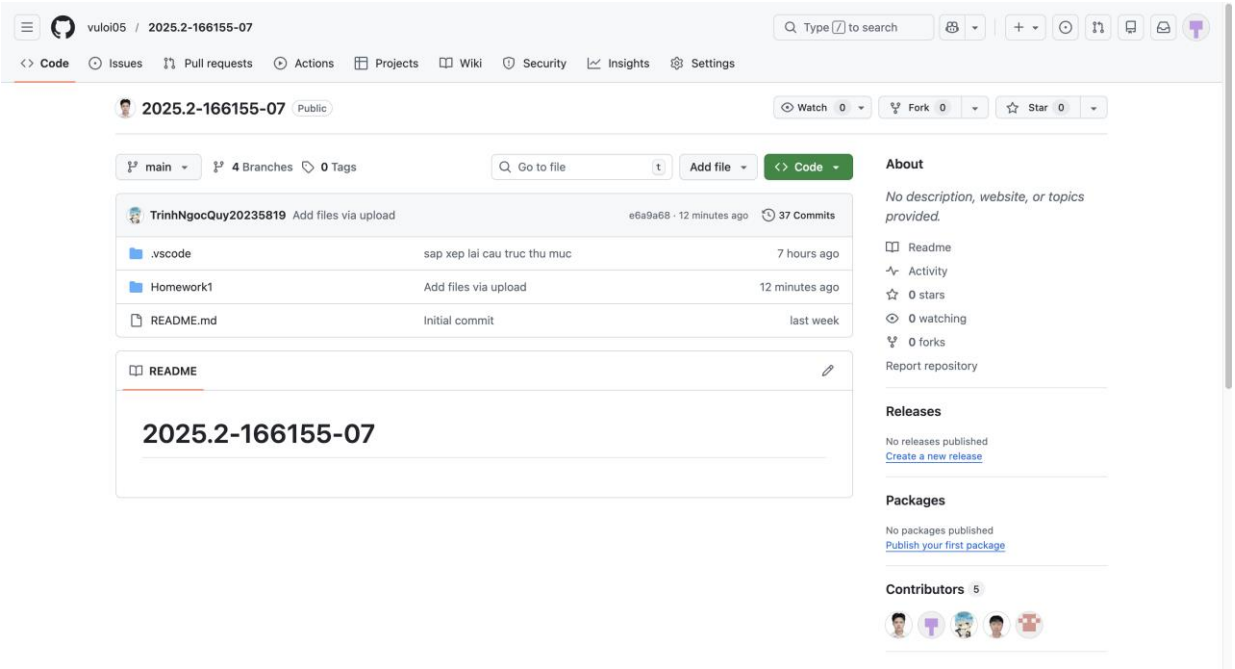
1 Giới thiệu

Bảng 1: Môi trường phát triển

Thành phần	Version
JDK	Java 24.0.1
JavaFX SDK	24.0.1
IDE	VS Code
OS	MacOS
Git	2.50.1

2 Thiết lập GitHub Repository



Link đến repository: github.com/vuloi05/2025.2-166155-07

3 Bài tập code

3.1 HelloWorld

Chương trình hỏi tên và hiển thị lời chào

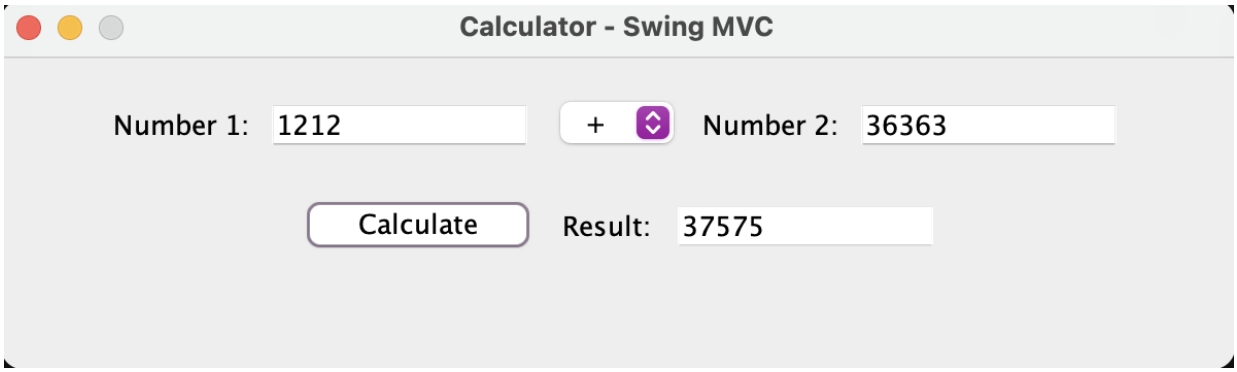
```java
import java.util.Scanner;

public class HelloWorld {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println(x: "Welcome to ITSS course!");

        System.out.print(s: "Please enter your name: ");
        String name = scanner.nextLine();

        System.out.println("Hello, " + name + "!");

        System.out.println(x: "This is from main working branch!");
        System.out.println(x: "This is from feature branch!");

        scanner.close();
    }
}
```

Kết quả:

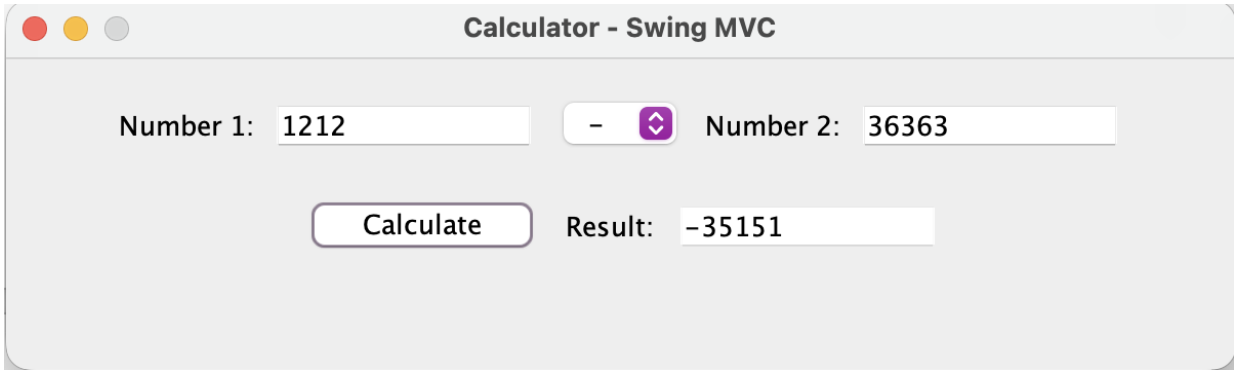
```
ductwan@MacBook-Pro-cua-Toan 20235846_NguyenDucToan % cd HelloWorld && /Library/Java/JavaVirtualMachines/jdk-24.jdk/Contents/Home/b
in/java -cp bin HelloWorld
Welcome to ITSS course!
Please enter your name: Toàn
Hello, Toàn!
This is from main working branch!
This is from feature branch!
ductwan@MacBook-Pro-cua-Toan HelloWorld %
```

3.2 Calculator Swing (MVC)

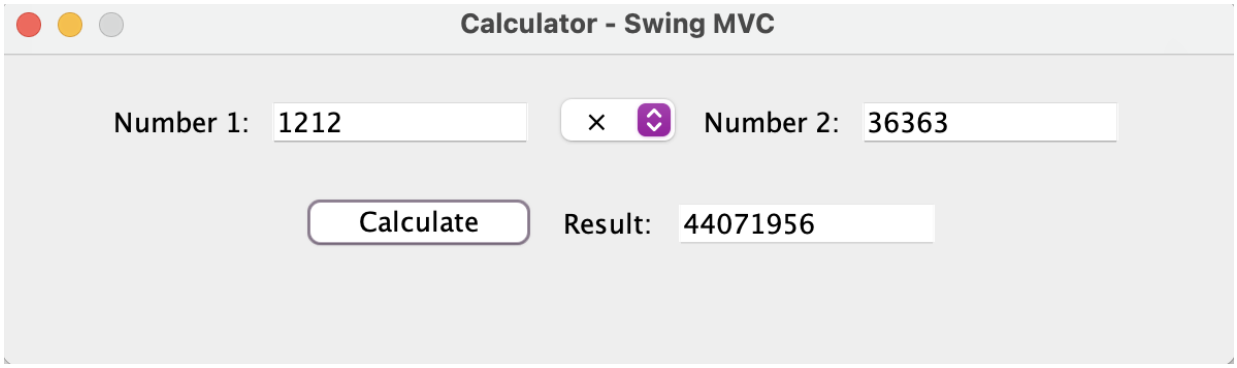
Máy tính 4 phép tính sử dụng Swing + MVC



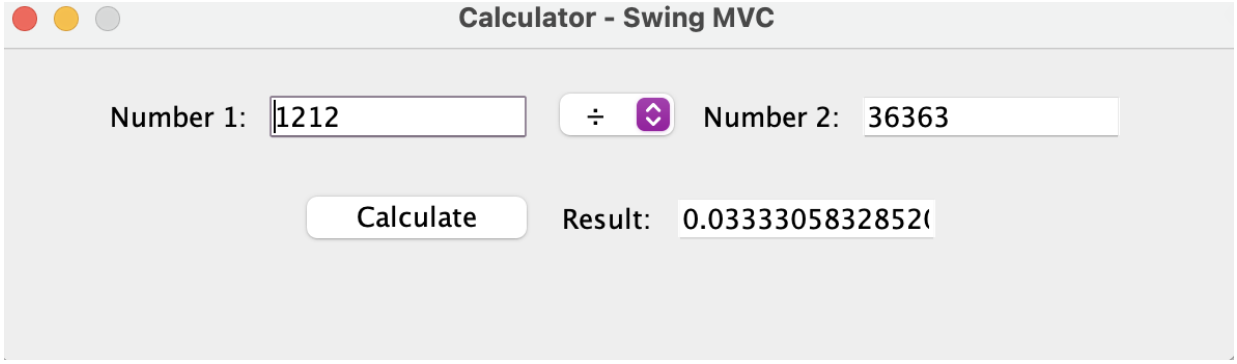
Hình 1: Phép cộng



Hình 2: Phép trừ



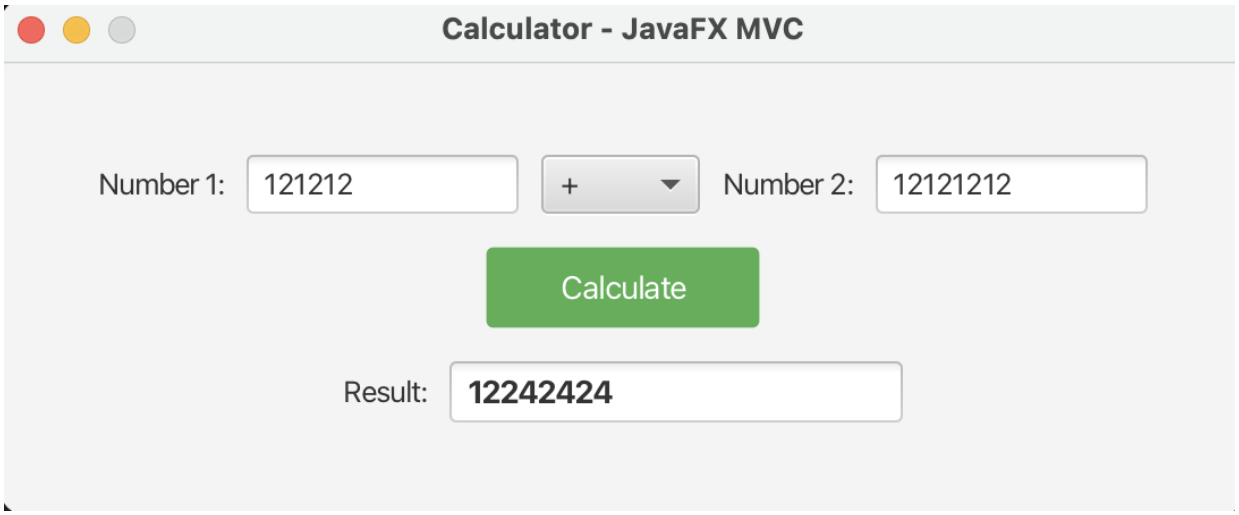
Hình 3: Phép nhân



Hình 4: Phép chia

3.3 Calculator JavaFX (MVC + Scene Builder)

Máy tính tương tự nhưng dùng JavaFX và FXML


Calculator - JavaFX MVC
Number 1:
121212
+
Number 2:
12121212
Calculate
Result:
12242424

4 Thao tác Git

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git add 20235846_NguyenDucToan
ductwan@MacBook-Pro-cua-Toan Homework1 % git status
On branch hw1/NguyenDucToan
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   20235846_NguyenDucToan/CalculatorJavaFX/bin/calculator/Main.class
        new file:   20235846_NguyenDucToan/CalculatorJavaFX/bin/calculator/view/calculator.fxml
        new file:   20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/Main.java
        new file:   20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/controller/CalculatorController.java
        new file:   20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/model/CalculatorModel.java
        new file:   20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/view/calculator.fxml
        new file:   20235846_NguyenDucToan/CalculatorSwing/bin/calculator/Main.class
        new file:   20235846_NguyenDucToan/CalculatorSwing/bin/calculator/controller/CalculatorController$CalculateListener.class
        new file:   20235846_NguyenDucToan/CalculatorSwing/bin/calculator/controller/CalculatorController.class
        new file:   20235846_NguyenDucToan/CalculatorSwing/bin/calculator/model/CalculatorModel.class
        new file:   20235846_NguyenDucToan/CalculatorSwing/bin/calculator/view/CalculatorView.class
        new file:   20235846_NguyenDucToan/CalculatorSwing/src/calculator/Main.java
        new file:   20235846_NguyenDucToan/CalculatorSwing/src/calculator/controller/CalculatorController.java
        new file:   20235846_NguyenDucToan/CalculatorSwing/src/calculator/model/CalculatorModel.java
        new file:   20235846_NguyenDucToan/CalculatorSwing/src/calculator/view/CalculatorView.java
        new file:   20235846_NguyenDucToan/HelloWorld/bin/HelloWorld.class
        new file:   20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
        new file:   20235846_NguyenDucToan/MVCTutorial/(1).classpath
        new file:   20235846_NguyenDucToan/MVCTutorial/(1).project
        new file:   20235846_NguyenDucToan/MVCTutorial/(2).classpath
        new file:   20235846_NguyenDucToan/MVCTutorial/.classpath
        new file:   20235846_NguyenDucToan/MVCTutorial/.project
        new file:   20235846_NguyenDucToan/MVCTutorial/MVC 3 implementations.txt
        new file:   20235846_NguyenDucToan/MVCTutorial/bin/CalculatorController$CalculateListener.class
        new file:   20235846_NguyenDucToan/MVCTutorial/bin/CalculatorController.class
        new file:   20235846_NguyenDucToan/MVCTutorial/bin/CalculatorModel.class
        new file:   20235846_NguyenDucToan/MVCTutorial/bin/CalculatorView.class
        new file:   20235846_NguyenDucToan/MVCTutorial/bin/MVCCalculator.class
        new file:   20235846_NguyenDucToan/MVCTutorial/src/CalculatorController.java
        new file:   20235846_NguyenDucToan/MVCTutorial/src/CalculatorModel.java
        new file:   20235846_NguyenDucToan/MVCTutorial/src/CalculatorView.java
        new file:   20235846_NguyenDucToan/MVCTutorial/src/MVCCalculator.java
```

Hình 5: Add file

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git commit -m "Add Homework1 for 20235846_NguyenDucToan"
[hw1/NguyenDucToan 57c00d1] Add Homework1 for 20235846_NguyenDucToan
 32 files changed, 729 insertions(+)
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorJavaFX/bin/calculator/Main.class
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorJavaFX/bin/calculator/view/calculator.fxml
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/Main.java
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/controller/CalculatorController.java
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/model/CalculatorModel.java
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorJavaFX/src/calculator/view/calculator.fxml
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/bin/calculator/Main.class
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/bin/calculator/controller/CalculatorController$CalculateListener.class
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/bin/calculator/controller/CalculatorController.class
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/bin/calculator/model/CalculatorModel.class
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/bin/calculator/view/CalculatorView.class
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/src/calculator/Main.java
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/src/calculator/controller/CalculatorController.java
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/src/calculator/model/CalculatorModel.java
 create mode 100644 Homework1/20235846_NguyenDucToan/CalculatorSwing/src/calculator/view/CalculatorView.java
 create mode 100644 Homework1/20235846_NguyenDucToan/HelloWorld/bin/HelloWorld.class
 create mode 100644 Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/(1).classpath
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/(1).project
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/(2).classpath
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/.classpath
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/.project
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/MVC 3 implementations.txt
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/bin/CalculatorController$CalculateListener.class
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/bin/CalculatorController.class
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/bin/CalculatorModel.class
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/bin/CalculatorView.class
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/bin/MVCCalculator.class
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/src/CalculatorController.java
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/src/CalculatorModel.java
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/src/CalculatorView.java
 create mode 100644 Homework1/20235846_NguyenDucToan/MVCTutorial/src/MVCCalculator.java
ductwan@MacBook-Pro-cua-Toan Homework1 %
```

Hình 6: Commit

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git push origin hw1/NguyenDucToan
Enumerating objects: 58, done.
Counting objects: 100% (58/58), done.
Delta compression using up to 12 threads
Compressing objects: 100% (44/44), done.
Writing objects: 100% (56/56), 17.00 KiB | 17.00 MiB/s, done.
Total 56 (delta 2), reused 43 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
remote:
remote: Create a pull request for 'hw1/NguyenDucToan' on GitHub by visiting:
remote:      https://github.com/vuloi05/2025.2-166155-07/pull/new/hw1/NguyenDucToan
remote:
To https://github.com/vuloi05/2025.2-166155-07.git
 * [new branch]      hw1/NguyenDucToan -> hw1/NguyenDucToan
ductwan@MacBook-Pro-cua-Toan Homework1 %
```

Hình 8: Push

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git diff
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class
index 089924d..229e437 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class
index 07f3d1b..d85d572 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class
index 512a97e..e8acc43 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class
index 4942dbd..d2de0bb 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class
:...skipping...
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class
index 089924d..229e437 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class
index 07f3d1b..d85d572 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class
index 512a97e..e8acc43 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class
index 4942dbd..d2de0bb 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class
index 4fdd507..6cc504a 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class and b/Home:...skipping...
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class
index 089924d..229e437 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController$CalculateListener.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class
index 07f3d1b..d85d572 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorController.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class
index 512a97e..e8acc43 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorModel.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class
index 4942dbd..d2de0bb 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/CalculatorView.class differ
diff --git a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class
index 4fdd507..6cc504a 100644
Binary files a/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class and b/Homework1/20235768_VuTienLoi/MVCTutorial/bin/MVCCalculator.class differ
diff --git a/Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java b/Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
index 49809b4..eef25fa 100644
--- a/Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
+++ b/Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
@@ -4,6 +4,8 @@ public class HelloWorld {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

+       System.out.println("Welcome to ITSS course!");
+
        System.out.print("Please enter your name: ");
        String name = scanner.nextline();

diff --git a/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst b/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-stat
us/maven-compiler-plugin/compile/default-compile/inputFiles.lst
index eae7faa..f64eb31 100644
--- a/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
+++ b/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
@@ -1,3 +1,3 @@
-D:\GitHub\2025.2-166155-07\Homework 01\calculator_JavaFX\src\main\java\controller\CalculatorController.java
```

```
@@ -4,6 +4,8 @@ public class HelloWorld {
     public static void main(String[] args) {
         Scanner scanner = new Scanner(System.in);

+        System.out.println("Welcome to ITSS course!");
+
         System.out.print("Please enter your name: ");
         String name = scanner.nextLine();

diff --git a/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst b/Homework1/TranVanMan
us/maven-compiler-plugin/compile/default-compile/inputFiles.lst
index eae7faa..f64eb31 100644
--- a/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
+++ b/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
@@ -1,3 +1,3 @@
-D:\GitHub\2025.2-166155-07\Homework 01\calculator_JavaFX\src\main\java\controller\CalculatorController.java
-D:\GitHub\2025.2-166155-07\Homework 01\calculator_JavaFX\src\main\java\main.java
-D:\GitHub\2025.2-166155-07\Homework 01\calculator_JavaFX\src\main\java\model\CalculatorModel.java
+/Users/ductwan/Developer/ITSS viet/2025.2-166155-07/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/src/main/java/main.java
+/Users/ductwan/Developer/ITSS viet/2025.2-166155-07/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/src/main/java/model/CalculatorModel.java
+/Users/ductwan/Developer/ITSS viet/2025.2-166155-07/Homework1/TranVanManh/Homework 01_Tran_Van_Manh/calculator_JavaFX/src/main/java/controller/CalculatorController.java
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~
(END)
```

Hình 9: Modify file

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git pull origin hw1/NguyenDucToan
From https://github.com/vuloi05/2025.2-166155-07
 * branch            hw1/NguyenDucToan -> FETCH_HEAD
Already up to date.
ductwan@MacBook-Pro-cua-Toan Homework1 %
```

Hình 7: Git Pull

```
ductwan@MacBook-Pro-cua-Toan Homework1 % touch 20235846_NguyenDucToan/temp.txt
ductwan@MacBook-Pro-cua-Toan Homework1 % git add 20235846_NguyenDucToan/temp.txt
ductwan@MacBook-Pro-cua-Toan Homework1 % git commit -m "Add temp file for testing"
[hw1/NguyenDucToan 641b03f] Add temp file for testing
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Homework1/20235846_NguyenDucToan/temp.txt
ductwan@MacBook-Pro-cua-Toan Homework1 %
ductwan@MacBook-Pro-cua-Toan Homework1 % git rm 20235846_NguyenDucToan/temp.txt
rm 'Homework1/20235846_NguyenDucToan/temp.txt'
ductwan@MacBook-Pro-cua-Toan Homework1 % git commit -m "Remove temp file"
[hw1/NguyenDucToan 2355fb7] Remove temp file
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 Homework1/20235846_NguyenDucToan/temp.txt
ductwan@MacBook-Pro-cua-Toan Homework1 %
```

Hình 10: Remove file

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git checkout -b feature/20235846-update
Switched to a new branch 'feature/20235846-update'
ductwan@MacBook-Pro-cua-Toan Homework1 %
```

Hình 11: Create branch

```
Automatic merge failed; fix conflicts and then commit the result.
ductwan@MacBook-Pro-cua-Toan Homework1 % git merge feature/20235846-update
Already up to date.
ductwan@MacBook-Pro-cua-Toan Homework1 % git add 20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
```

Hình 12: Gộp nhánh

```
ductwan@MacBook-Pro-cua-Toan Homework1 % git merge feature/20235846-update
Auto-merging Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
CONFLICT (content): Merge conflict in Homework1/20235846_NguyenDucToan/HelloWorld/src/HelloWorld.java
Automatic merge failed; fix conflicts and then commit the result.
ductwan@MacBook-Pro-cua-Toan Homework1 %
```

Hình 13: Gộp nhánh có lỗi - 1

```
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes | Resc
<<<<<<< HEAD (Current Change)
        System.out.println(x: "This is from main working branch!");
=======
        System.out.println("This is from feature branch!");
>>>>>>> feature/20235846-update (Incoming Change)

        scanner.close();
    }
}
```

Hình 14: Gộp nhánh có lỗi - 2

```
        System.out.println(x: "This is from main working branch!");
        System.out.println(x: "This is from feature branch!");

        scanner.close();
    }
}
```

Hình 15: Sửa lỗi gộp nhánh